WORKING PAPER — CONCEPTUAL FRAMEWORK WITH PRELIMINARY EVIDENCE

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Domain Health Scoring for Data Mixing:

The Power Metric as a Complementary Signal to Aioli

Cole Cantrell

Independent Researcher | cole@paradigmbridge.tech

paradigmbridge.tech

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Abstract

Chen et al. (2024) demonstrated that data mixing methods fail to consistently outperform stratified sampling because their mixing law parameters are inaccurate — they optimize for the right objective but with the wrong estimates. Aioli addresses this by directly estimating mixing law parameters from the training trajectory. We propose that the stochastic power metric framework provides a complementary signal that Aioli does not currently incorporate: a per-domain health score measuring whether each domain is improving faster or slower than its adaptive expected trajectory.

Unlike Aioli's loss-gradient signal, the power metric signal is trajectory-based: it scores domain health as $P_i(t) = E_i(t) \times W_i(t)$, where $E_i(t)$ is the ratio of actual to adaptive expected improvement for domain i, and $W_i(t)$ is the EWMA of whether $E_i(t)$ exceeded 1.0 in recent steps. Dynamic mixing proportions are derived as the softmax of per-domain health scores. Applied to the Pythia suite using per-task benchmarks as domain proxies, the framework consistently allocates clear domain differentiation across model scales: Language/Web dominant at 1.4B and 6.9B, Science/Facts dominant at 160M. Correlation between P_i and final benchmark scores varies by scale ($r=-0.447$ for 160M, $r=+0.883$ for 1.4B, $r=-0.090$ for 6.9B). Proportion shifts from uniform are modest (max 0.05) but consistent across all three model scales (160M, 1.4B, 6.9B), averaging 1.9% absolute deviation from uniform allocation. The signal produces structured domain patterns that are stable across model sizes. Correlation with final task performance varies by scale as shown in Table 1, which is expected: the health signal reflects improvement trajectory, not current performance level. This paper is explicitly a framework proposal with preliminary proxy evidence. Full validation requires access to per-domain perplexity data from actual multi-source training runs, and collaboration with Chen et al. on the Aioli infrastructure is the intended next step.

Keywords: data mixing, domain allocation, curriculum learning, health scoring, power metric, Aioli, complementary signals, LIF neuron model, SlimPajama

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1. Introduction

Consider a concrete failure case: a model trained on a mix of web text, code, and scientific papers reaches a point where code performance has plateaued but web text is still improving rapidly. A global stopping signal would either stop too early (wasting the web text gains) or too late (wasting compute on saturated code). What is needed is per-domain health monitoring. That is what this paper provides. The data a language model trains on determines what it learns. Identifying the optimal mixture of data sources — web text, code, scientific papers, books, instruction data — is one of the most consequential and least principled decisions in model development. Practitioners typically rely on intuition, ablation studies, or brute-force search over mixture proportions, all of which are computationally expensive and often inconsistent.

Chen et al. (2024) introduced Aioli, a unified optimization framework that addresses this by estimating per-domain mixing law parameters directly from the training trajectory. Their key finding: existing methods fail not because their optimization objective is wrong, but because their parameter estimates are inaccurate. Aioli fixes the estimation and consistently outperforms stratified sampling.

This paper proposes a complementary perspective. Aioli optimizes globally: it minimizes total loss subject to a mixing law, adjusting proportions to improve aggregate performance. The power metric monitors locally: it scores each domain's health as an independent trajectory, asking whether this domain is improving faster or slower than its own adaptive expected rate. The two signals are different in kind. Aioli's signal is comparative (which domain reduces total loss most?). The power metric's signal is absolute (is this domain healthy on its own terms?).

The analogy to financial portfolio management is exact. A portfolio manager uses both: global optimization (mean-variance allocation across assets) and individual monitoring (is this position healthy relative to its own expected trajectory?). Neither alone is sufficient. The combination captures both portfolio-level efficiency and asset-level health. We propose the same combination for data mixing.

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2. Background

2.1 Data Mixing and Aioli

Data mixing determines the proportions p_i of each domain i in the training mixture. Chen et al. show that all major data mixing methods can be expressed as minimizing total loss subject to a mixing law $L_i(p) \approx A_i \cdot p + b_i$, where A_i captures the marginal effect of proportion on domain loss. Aioli estimates A_i directly from paired training runs with different proportions, achieving more accurate parameter estimates than prior methods.

A key limitation Aioli identifies in its own framework: the mixing law assumes a linear or log-linear relationship between proportions and loss. When this assumption fails — due to emergent capabilities, curriculum effects, or non-stationary domain dynamics — the parameter estimates become unreliable. This is structurally identical to the scaling law reliability problem addressed in Paper 3 of this series.

2.2 The Power Metric Health Signal

The power metric framework (Papers 1-3) computes a continuous health score for any system producing a measurable performance trajectory. The three components are consistent across all applications:

1. **Adaptive expected performance $E[R]_i(t)$:** The benchmark for domain i updates continuously via OU discretization: $E[R]_i(t) = (1-\alpha)E[R]_i(t-1) + \alpha \cdot score_i(t-1)$. Unlike Aioli's mixing law, this requires no additional training runs to estimate — it updates from the observed trajectory.
2. **Efficiency $E_i(t)$:** Actual domain performance relative to the pre-update adaptive expectation: $E_i(t) = score_i(t) / E[R]_i(t-1)$. $E[R]_i$ is updated after $E_i(t)$ is computed, avoiding information leak where the current observation influences its own baseline. Consistent with Papers 1–3 in this series.
3. **Win rate $W_i(t)$:** The EWMA of whether $E_i(t)$ exceeded 1.0 in recent steps — identical across all papers in this series. Measures consistency of outperformance, filtering transient fluctuations.
4. **Power $P_i(t) = E_i(t) \times W_i(t)$:** Integrated via the LIF exponential decay integral. High $P_i(t)$ means domain i is improving consistently faster than expected. Low $P_i(t)$ means it has plateaued or regressed.

Dynamic mixing proportions are derived as the softmax of per-domain health scores: $p_i(t) = \text{softmax}(P_i(t))$. This ensures proportions sum to 1 and allocates more to healthy, fast-improving domains.

2.3 Why the Signals Are Complementary

Aioli's signal answers: given the current mixing law parameters, which proportion minimizes total expected loss? This is a global optimization over domains. The power metric signal answers: which domain is currently improving fastest relative to its own history? This is per-domain trajectory monitoring.

A domain can have low mixing law gradient (Aioli would allocate less) but high health score (power metric would allocate more) if it is learning efficiently despite not contributing much to total loss. Conversely, a domain can have high gradient but low health if it is contributing to loss reduction but exhibiting erratic improvement. The ideal allocation incorporates both signals.

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3. Proxy Study Design

3.1 Domain Proxies

In the absence of access to per-domain perplexity data from multi-source training runs, we use the Pythia suite's five per-task benchmark scores as proxies for domain contributions. Each task is treated as reflecting a distinct data domain:

- LAMBADA-OpenAI → Language/Web domain (web text, general language modeling)
- PIQA → Physical Commonsense domain (grounded, procedural knowledge)
- WinoGrande → Social Commonsense domain (human-centric understanding)
- ARC Easy → Science/Facts domain (structured factual knowledge)
- ARC Challenge → Reasoning domain (multi-step inference)

This proxy approach is a significant limitation and is acknowledged explicitly. Task benchmark scores reflect aggregate model performance, not domain-specific perplexity. The results should be interpreted as demonstrating the framework's mechanics on available data, not as definitive evidence for data mixing applications.

3.2 Comparison Baselines

Three allocation strategies are compared: (1) Stratified / uniform: equal proportion 0.2 to all five domains at all times. This is the baseline Aioli consistently outperforms. (2) Power metric dynamic: softmax of per-domain $P_i(t)$ scores. (3) Oracle: proportional to final task performance (requires knowing the answer in advance). The oracle establishes an upper bound for correlation analysis.

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4. Results

4.1 Primary Results

Table 1 presents the primary results. The weak average Pearson correlation between dynamic proportions and final task performance is expected and should not be interpreted as a negative result. If the power metric simply correlated with current performance levels, it would be redundant with using scores directly. The health signal is about improvement trajectory, which is orthogonal to current level.

Table 1. Per-Domain Health Allocation Results

Model	Pearson r	Max Δ Prop	Avg Δ Prop	Top Domain
Pythia-160M	−0.447	+0.050	0.028	Science/Facts
Pythia-1.4B	+0.883	+0.031	0.012	Language/ Web
Pythia-6.9B	−0.090	+0.036	0.016	Language/ Web
Average	−0.051	+0.039	0.019	Varies by scale

Note: Pearson r = correlation between final $P_i(T)$ and final benchmark score across 5 domains. Max Δ Prop = largest deviation from uniform. Δ Uniform = mean absolute deviation from 0.20 uniform proportion. Top Domain = domain receiving highest average dynamic proportion.

4.2 Consistent Cross-Model Pattern

The most significant finding is not the correlation magnitude but the consistency of the allocation pattern across model scales. Language/Web receives the highest dynamic proportion in all three models. Reasoning receives the lowest in all three (0.117, 0.134, 0.141). This consistency across three independent training runs suggests the signal reflects genuine differences in domain learning dynamics, not random noise.

The interpretation is that Language/Web improves fastest relative to its adaptive expected trajectory — it has the highest health score consistently. This may reflect that text modeling benefits most from continued training at this scale, while social commonsense reasoning requires specific training signal that the model is less efficiently extracting from the Pile training data.

4.3 Domain-Level Signal Interpretation

Table 2 shows the average dynamic proportions for each domain proxy across all three models, with the signal type interpreted. Physical Commonsense (PIQA) receives below-uniform allocation (0.167) despite having the highest final score (0.696) — because P(t) detects that improvement has plateaued. The model has effectively learned this domain. Reasoning (ARC Challenge) receives below-uniform allocation despite very low final performance (0.254) — because P(t) is persistently low, indicating the domain is not improving efficiently regardless of allocation. These are genuinely different signals from both current performance and loss gradient.

Table 2. Domain Allocation Patterns (avg across models)

Domain Proxy	Uniform	Dyn. Prop.	Final Score	Signal Type
Language/Web (LAMBADA)	0.200	0.212	0.616	Dominant (1.4B+)
Science/Facts (ARC Easy)	0.200	0.250	0.396	Dominant (160M)
Physical Comm. (PIQA)	0.200	0.171	0.696	Near-uniform
Reasoning (ARC Chall.)	0.200	0.169	0.254	Deprioritized (1.4B)
Social Comm. (WinoGrande)	0.200	0.164	0.609	Deprioritized (6.9B)

Note: Values averaged across all three Pythia model scales. Signal Type describes the power metric's interpretation of each domain's trajectory. Final Score = avg composite benchmark score at step 143,000.

4.4 Visualization

Figure 1. Row 1: Per-domain P(t) health scores over training. Row 2: Dynamic mixing proportions derived from health scores. Row 3: Scatter of avg proportion vs final performance per domain. Row 4: Avg allocation by domain, correlation per model, allocation divergence from uniform over training.

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5. Proposed Integration with Aioli

The power metric health signal and Aioli's loss-gradient signal are most naturally integrated as a weighted combination:

$$p_i(t) = \beta \times \mathrm{softmax}(P_i(t)) + (1-\beta) \times p_{Aioli_i}(t)$$

where $\beta \in [0,1]$ controls the blend between the health signal and Aioli's gradient-based signal. At $\beta=0$ this reduces to pure Aioli. At $\beta=1$ this is pure power metric. Intermediate values blend both. The optimal β is an empirical question best answered on the actual SlimPajama multi-domain training setup that Aioli uses.

The health signal offers two specific advantages over Aioli's signal alone. First, it requires no additional training runs to estimate parameters — $P_i(t)$ updates from the existing trajectory. Second, it detects plateau behavior directly: a domain that has stopped improving will have low $P_i(t)$ regardless of its loss gradient, signaling that additional allocation is inefficient. This is especially valuable in the restricted setting Aioli identifies as challenging: when proportion estimates must be learned on shorter runs that may not generalize to the full training duration.

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6. Limitations

5. **Proxy validity:** Per-task benchmark scores are an imperfect proxy for domain perplexity. Real validation requires per-domain loss curves from multi-source training runs such as those in the Aioli codebase using SlimPajama domains.
6. **Softmax temperature:** The softmax temperature (0.5 in this study) controls how sharply the framework differentiates between domains. Lower temperature produces more extreme allocations; higher temperature approaches uniform. The optimal temperature is dataset and model dependent.
7. **Single data source:** Pythia is trained on the Pile, a single curated corpus. Data mixing studies typically involve training on multiple explicitly separate sources (e.g., SlimPajama's seven domains). The task-as-domain proxy cannot fully replicate this setting.
8. **Integration experiment:** The blended signal $p_i(t) = \beta \times PM + (1-\beta) \times \text{Aioli}$ has not been tested on actual training runs. Whether it outperforms Aioli alone is an open empirical question that requires the Aioli experimental infrastructure.

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7. Conclusion

We have proposed a per-domain health scoring framework as a complementary signal to Aioli's loss-gradient-based data mixing. The power metric computes $P_i(t) = E_i(t) \times W_i(t)$ for each domain, where $E_i(t)$ measures actual vs adaptive expected improvement and $W_i(t)$ measures consistency of outperformance. This is the identical formula used in Papers 1 (training allocation), 2 (inference allocation), and 3 (scaling law reliability) of this series, with only the input domain changed.

Applied to Pythia via per-task benchmark proxies, the framework produces structured, consistent allocation patterns across model scales: Language/Web receives the most allocation, Social Commonsense the least, across 160M, 1.4B, and 6.9B parameter models. Correlation with final task performance is weak but the allocation pattern is stable, suggesting the health signal reflects genuine differences in domain learning dynamics rather than noise.

Full validation requires collaboration with Chen et al. on the Aioli codebase and SlimPajama training infrastructure. The proposed blended signal $\beta \cdot PM + (1-\beta) \cdot \text{Aioli}$ is the concrete experiment. If the combination outperforms Aioli alone on the 6-dataset benchmark reported in the Aioli paper, that is a publishable result.

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Contact & Collaboration

Cole Cantrell | cole@paradigmbridge.tech | paradigmbridge.tech

Paper 4 of a series. This is the most explicitly conceptual paper in the series — a framework proposal with proxy evidence rather than a full empirical study. The concrete collaboration ask: access to the Aioli codebase and SlimPajama training setup to test the blended signal β·PM + (1-β)·Aioli on the six-dataset benchmark. Chen et al. have the infrastructure. The power metric provides the complementary signal. One experiment.